Math-1151 Class Test 02 Section A

1. Sketch the area represented by $\boldsymbol{f(x)} = \begin{cases} \mathbf{5}, & \mathbf{0 \le x \le 2} \\ \mathbf{3 + x}, & \mathbf{2 < x \le 4} \end{cases}$ for $\int_{\mathbf{0}}^{\mathbf{4}} \boldsymbol{f(x)\,dx}$. Evaluate the integral using Fundamental theorem of Calculus. **[4]**

2. Graph the function $\boldsymbol{f(x) = x + 3}$ over the interval $[\mathbf{-3}\,, \boldsymbol{x}]$. Then use simple area formula from geometry to find the area function $\boldsymbol{A(x)}$ that gives the area under the graph and confirm that $\boldsymbol{A'(x) = f(x)}$. **[4]**

3. According to following figure evaluate $\int_{-7}^{8} f(x)dx$. **[2]**